



MSc program in Business Administration and Data Science
Department of Digitalization

**PREDICTING CUSTOMER CHURN: A MACHINE
LEARNING FRAMEWORK WITH REVENUE-AT-RISK
QUANTIFICATION**

Quantitative Risk Management - An Application of Machine Learning (CPOLV2506U)

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Number of pages: 15

Word count: 5.980

Submission date: 12-12-2025

Abstract

This study predicts customer churn using supervised machine learning models and quantifies the associated financial exposure through the Revenue-at-Risk (RaR) framework applied to a customer portfolio. The problem addressed is whether Logistic Regression, Random Forest, and XGBoost classifiers can effectively identify churn-prone customers while supporting risk-adjusted revenue estimation. The modelling workflow incorporates data preprocessing, feature engineering, dimensionality reduction using PCA, customer clustering using K-Means, class balancing with SMOTE, supervised modelling, and model interpretability through SHAP values. Findings show that the models perform very similarly, but ensemble methods minimise false negatives. The Random Forest model obtains the highest class-specific recall (0.89) minimizing best the misclassified churned customers with similar accuracy level (0.83) and maintaining a F1-score comparable to the alternative models. Additionally, K-Means identifies a targetable cluster collecting 68,1% of total churned customers in the dataset, and SHAP analysis identifies the number of technical support tickets, charged amount, and customer tenure as the most important features explaining churn.

Keywords: Customer Churn Prediction; Revenue-At-Risk (RaR); Classification Models; Clustering; SMOTE; SHAP Values; AI Governance; Quantitative Risk Management.

List of Abbreviations

Abbreviation	Definition
ARR	Annual Recurrent Revenue
PD	Probability-of-Default
LGD	Loss-Given-Default
EDA	Exploratory Data Analysis
NaN	Not a Number
PCA	Principal Component Analysis
SMOTE	Synthetic Minority Over-Sampling Technique
ROC-AUC	Receiver Operating Characteristic - Area Under the Curve
SHAP	SHapley Additive Planations

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1 Introduction

Customer churn is one of the major business risks for subscription-based businesses, particularly in the telecommunications sector where high customer acquisition costs make retention key to mitigate revenue volatility. From a risk management perspective, each active customer contributes an exposure whose loss materialises when the customer discontinues the service. Modelling this enables a quantitative assessment of churn likelihood and the resulting financial impact, and mitigates the risks embedded in the customer portfolio.

Building on this perspective, the study first applies unsupervised techniques to cluster customers and identify behavioural groupings, and subsequently employs supervised machine learning models to classify customers as churn or non-churn. The resulting predicted probabilities are then integrated into a Revenue-at-Risk (RaR) framework that quantifies the expected monetary loss associated with potential future churn. The objective is to identify risk-bearing customer segments, develop an effective classification model, and estimate the associated financial impact while assessing how well different models detect high-risk cases.

2 Theoretical Framework

Risk often denotes a numerical representation of severity that is obtained by multiplying the probability of an unwanted event with a measure of its disvalue (Hansson 2004). Churn exhibits these characteristics, combining both the likelihood that a customer will discontinue the service and the financial impact associated with the loss of recurring revenue, this makes churn a central risk factor for subscription-based firms. The collapse of MoviePass illustrates this danger: after reducing its subscription price to rapidly scale, the firm experienced severe cash burn and high churn among unprofitable users, ultimately undermining the viability of its model (Esty and Fisher 2019).

Credit risk modelling commonly decomposes expected loss into the *probability-of-default* (PD), *loss-given-default* (LGD), and *exposure-at-default* (EAD)), formalised as:

$$EL = PD \times LGD \times EAD$$

PD captures the likelihood of a default event, whereas LGD and EAD quantify the financial impact conditional on default (Basel Committee on Banking Supervision 2005). A similar structure can be applied to churn prediction by adopting a Value-at-Risk perspective, where financial exposure is assessed by estimating potential losses under adverse conditions. This study adapts these concepts to churn by treating the model-estimated churn probability as the PD analogue and the customer's annualised revenue as the LGD–EAD component, defining a Revenue-at-Risk (RaR) metric that quantifies churn risk in monetary terms.

Machine learning is widely used to address churn risk, with classification models remaining the standard approach for predicting whether customers are likely to defect. Even though more complex deep learning models have emerged, non-neural network methods continue to be widely applied due to their strong performance and practical usability. In the churn prediction tournament analysed by Neslin et al. 2006, 44 analytics teams developed churn models on the same dataset using diverse modelling strategies. Tree-based models, particularly boosted decision trees, achieved the highest predictive accuracy but the authors also note that, interpretability also plays a crucial role in mitigating this risk. As K. Peng, Y. Peng, and Li 2023 introduced, boosting methods suffer from limited interpretability, and SHAP framework is introduced to decompose predictions into feature-level contributions and add an explanatory layer to the model.

3 Case Background and Data Description

The telecommunications industry is characterized by subscription-based revenues, translated into stable income that is usually measured with Annual Recurring Revenue (ARR). It is also characterized by high acquisition costs, which makes customer retention a critical determinant of success in this business. Tiny variations in churn rates can affect ARR predictability, cash flow, and long-term firm value. From a risk management perspective, customer churn represents a measurable form of portfolio risk since each active customer constitutes an exposure whose loss materializes when the contract is discontinued.

Using the Basel II framework explained in Section 2, this study models churn as a *probability-of-default* (PD) event within a customer portfolio, where the resulting revenue shortfall corresponds to the *loss-given-default* (LGD). The objective is to identify key drivers of churn, estimate churn probabilities at the individual level, and quantify the associated expected loss to support proactive risk mitigation. The analysis applies machine learning techniques to estimate these probabilities and evaluate the resulting revenue-at-risk within a simulated telecom portfolio. It proceeds by first exploring customer behavior and identifying latent risk concentrations through exploratory data analysis (EDA), followed by the development of classification models that estimate the likelihood of churn for each customer.

The case uses an open-access dataset offered by *Kaggle*, representing a synthetic population of telecom subscribers. Each observation corresponds to an individual customer account, containing a wide range of variables (See Table 1) including demographic variables (e.g., gender, seniority), service characteristics (e.g., contract type, internet service, payment method), and numerical measures of engagement (e.g., tenure, monthly charges, total charges, and number of technical or administrative tickets).

Variable	Type (Group)	Description
customerID	Identifier	Unique identifier for each customer.
gender	Demographic	Customer gender.
SeniorCitizen	Demographic	Indicates whether the customer is a senior citizen (1) or not (0).
Partner	Demographic	Whether the customer has a partner.
Dependents	Demographic	Whether the customer has dependents.
PhoneService	Service characteristic	Indicates if the customer has a phone service.
MultipleLines	Service characteristic	Whether the customer has multiple phone lines.
InternetService	Service characteristic	Type of internet service (DSL, Fiber optic, None).
OnlineSecurity	Service characteristic	Indicates if online security is included.
OnlineBackup	Service characteristic	Indicates if online backup is included.
DeviceProtection	Service characteristic	Indicates if device protection is included.
TechSupport	Service characteristic	Indicates if technical support is included.
StreamingTV	Service characteristic	Whether the customer has streaming TV.
StreamingMovies	Service characteristic	Whether the customer has streaming movies.
Contract	Service characteristic	Contract type (Month-to-month, One-year, Two-year).
PaperlessBilling	Service characteristic	Whether billing is paperless.
PaymentMethod	Service characteristic	Customer payment method.
MonthlyCharges	Engagement measure	Monthly amount charged to the customer.
tenure	Engagement measure	Number of months the customer has stayed with the company.
TotalCharges	Engagement measure	Total amount charged to date.
numAdminTickets	Engagement measure	Number of administrative support tickets opened.
numTechTickets	Engagement measure	Number of technical support tickets opened.
Churn	Target variable (churn)	1 if the customer discontinued service, 0 otherwise.

Table 1: Raw dataset overview (variables, types, and brief definitions)

Each observation includes the target variable Churn, which equals 1 if the customer discontinued service within the observed period and 0 otherwise. This variable is the one predicted and modeled. The initial dataset contains 7.043 observations and 23 features and will be transformed in Section 4.1 and 4.2.

4 Methodology

The study follows the methodology approach shown in Figure 1. The workflow begins with data preprocessing, where irrelevant features and inconsistent entries are removed, and data types are standardized. The following step is exploratory data analysis (EDA), which provides an overview of the dataset and identifies potential customer clusters with similarities. Data preparation involves feature

engineering, addressing class imbalance, and creating train–test splits with appropriate scaling. The process then moves to model configuration, where suitable algorithms are selected, and evaluation metrics are defined. Finally, the models are trained and their performance assessed to identify the most effective approach for predicting customer churn.

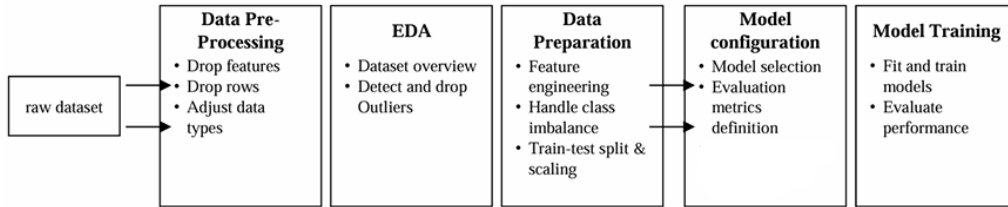


Figure 1: End-to-end modelling pipeline

4.1 Data Preprocessing

Data cleaning and preparation are key when applying classification machine learning models, as many of them only accept numerical inputs. This process was performed using python. The dataset comprised 7,043 observations (customers) with 23 explanatory variables. An initial inspection of the data revealed information stored in multiple formats (see Figure 2), such as binary variables expressed both as 0/1 and as True/False for the same type of feature. Furthermore, the column data types were limited to two formats: integers and objects. For subsequent modeling, all data must be numeric; an object column indicates that the variable is stored as a string or contains inconsistent entries combining numeric and non-numeric values.

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...
4	9237-HQ/TU	Female	0	No	No	2	Yes	No	Fiber optic	No	...
	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	numAdminTickets	numTechTickets	Churn		
	No	Month-to-month	Yes	Electronic check	29,85	29,85	0	0	No		
	No	One year	No	Mailed check	56,95	1889,5	0	0	No		
	No	Month-to-month	Yes	Mailed check	53,85	108,15	0	0	Yes		
	No	One year	No	Bank transfer (automatic)	42,3	1840,75	0	3	No		
	No	Month-to-month	Yes	Electronic check	70,7	151,65	0	0	Yes		

Figure 2: First 5 rows of the raw data before preparation

Before addressing the different data formats, missing values were checked, and none resulted in NaN. Since the dataset contains objects, a sanity check was also performed for empty strings (""), that are not detected by the NaN check; five rows were found in binary columns and were assumed to correspond to zeros. Once the dataset was complete, data arrangement could be performed. First, a

formatting error was identified in the `MonthlyCharges` and `TotalCharges` columns, where decimal and thousand separators were mixed (e.g., “1.234,56”). The dot thousand separators were removed, and commas were replaced with dots. Finally, these two columns were converted to integers.

4.2 Data Preparation and EDA

Once the data was in the proper format and consistent form, preparing it into a format that models can understand is essential. Starting with the `tenure` variable, it was bucketized into the following month ranges: 0-6, 7-12, 13-24, 25-36, 37-48, 49-60, and 61+. This transformation is important in classification models because it captures the nonlinear relationship between customer tenure and churn probability, enhancing model interpretability by grouping customers into clearly defined risk segments.

Feature encoding was also performed to ensure model robustness, since all variables must be in a similar format to avoid bias between features. The variable `gender` was converted into a binary indicator named `male`, where 1 represents male and 0 female. All binary variables with Yes/No responses, such as `Partner`, `Dependents`, `PaperlessBilling`, `Churn`, and `PhoneService`, were transformed into numerical 0/1 format. Variables that included the category No internet service, such as `OnlineSecurity`, `OnlineBackup`, `DeviceProtection`, `TechSupport`, `StreamingTV`, and `StreamingMovies`, were also encoded consistently as binary values. Likewise, `MultipleLines` was converted to binary by assigning 0 to the categories No and No phone service, and 1 to Yes.

For categorical variables with more than two categories, namely `InternetService`, `Contract`, and `PaymentMethod`, one-hot encoding was applied using `pd.get_dummies()` while dropping the first category to prevent multicollinearity. Additionally, all Boolean variables were converted to numeric values, and spaces in column names were replaced with underscores for consistency. Finally, no variables were dropped, as all were considered potentially informative for churn classification. After the initial preprocessing steps, the dataset contained 7.043 observations with 26 features, see Figure 3.

	SeniorCitizen	Partner	Dependents	PhoneService	MultipleLines	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	...
0	0	1	0	0	0	0	1	0	0	0	...
1	0	0	0	1	0	1	0	1	0	0	...
2	0	0	0	1	0	1	1	0	0	0	...
3	0	0	0	0	0	1	0	1	1	0	...
4	0	0	0	1	0	0	0	0	0	0	...
Churn	tenure_group	male	InternetService_Fiber_optic	InternetService_No	Contract_One_year	Contract_Two_year	PaymentMethod_Credit_card_(automatic)				
0	1	0	0	0	0	0	0				
0	4	1	0	0	1	0	0				
1	1	1	0	0	0	0	0				
0	5	1	0	0	1	0	0				
1	1	0	1	0	0	0	0				

Figure 3: First 5 rows of the data after preparation

Histograms served as a useful tool to understand the structure of the features, as they can reveal spe-

cific patterns, class imbalances, or the presence of outliers. As seen in Figure 10 in the Appendix, `numAdminTickets` and `numTechTickets` exhibit an almost identical distribution, with most observations concentrated at 0. These variables are likely related to churn, a hypothesis confirmed by the correlation of 0.357 between `numTechTickets` and `Churn` (see Figure 11 in the Appendix), indicating that customers who open more technical support tickets are more likely to churn. No strong patterns or anomalies are observed in the remaining numerical variables, see Figure 10.

Binary distributions, shown in Figure 9 in the Appendix, provide a clearer understanding of the dataset’s composition. For instance, the `SeniorCitizen` distribution indicates that the sample consists predominantly of younger customers, while `Partner` shows a balanced proportion between customers with and without a partner. The `Gender` variable is also evenly distributed. Additionally, `MultipleLines` reveals that most customers have a single contracted line, although a considerable share possesses multiple lines. The distributions of `StreamingTV` and `StreamingMovies` display a similar pattern to `MultipleLines`. Finally as observed in Figure 4, `Churn` exhibits a clear class imbalance toward non-churned customers (0), which will be further discussed in Section 4.3.

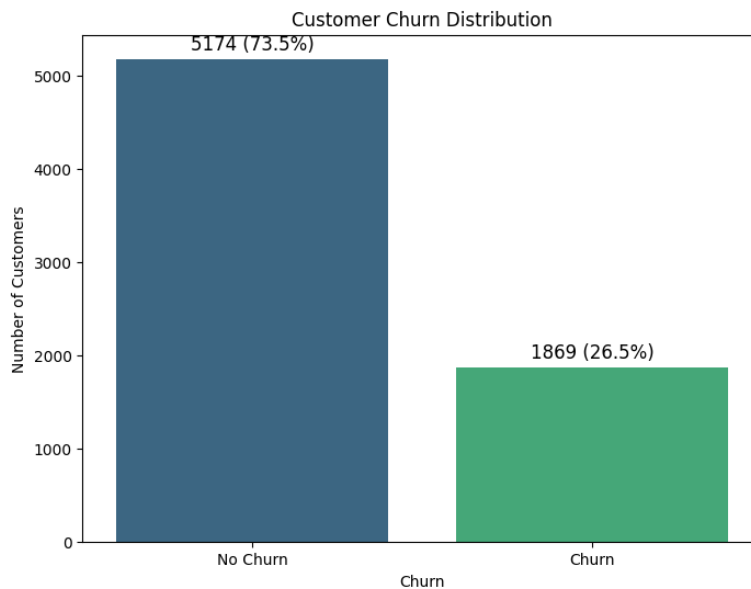


Figure 4: Churn variable distribution before balancing

Since the objective of this project is to identify customers prone to churn, a clustering approach was applied to detect common behavioral patterns among customers. K-Means clustering was selected for its efficiency and interpretability, as it partitions observations into k groups by minimizing within-cluster variance. Given that the dataset contains 26 features, Principal Component Analysis (PCA) was applied to reduce dimensionality, remove redundancy, and retain most of the variance. PCA identifies the axis that accounts for the largest amount of variance in the training set, the principal components (Géron 2017). This clustering approach was performed without the `Churn` variable to

isolate the effect of this one during the clustering process. The Elbow method, which evaluates the marginal gain in explained variance as k increases, was then used to determine the optimal number of clusters (see Figure 12 in the Appendix), with $k = 3$ providing the best trade-off between model simplicity and explanatory power.

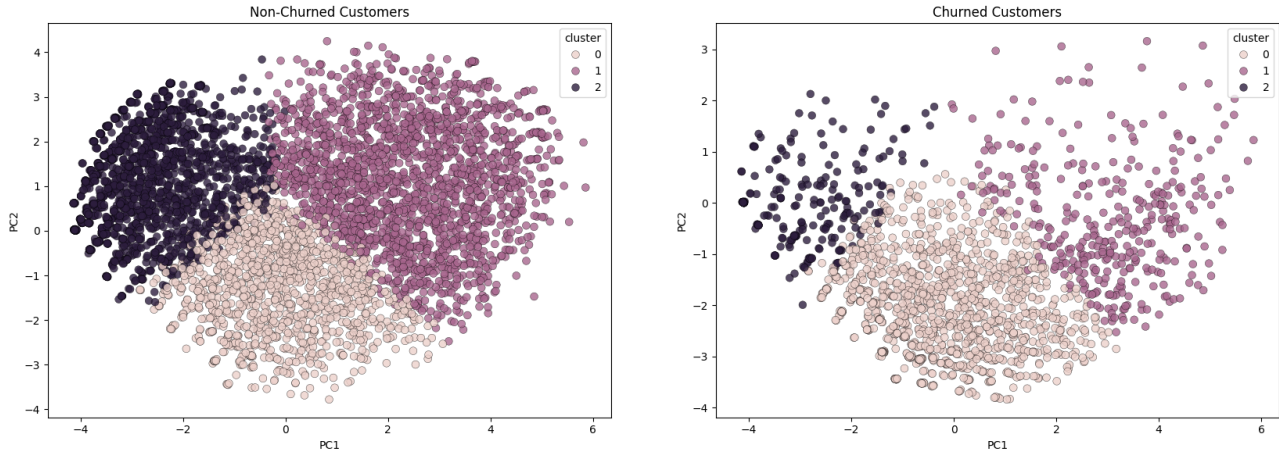


Figure 5: Non-churned and churned customers across clusters on the first two PCs

As seen in Figure 5, the three clusters are clearly defined along the PC1 and PC2 axes, where it can be visually observed that Cluster 0 concentrates most of the churned customers. As observed in Table 2, a total of 1,272 out of 1,869 churned customers (68.1%) are concentrated in Cluster 0 despite being the smallest cluster. Half of the customers in Cluster 0 have churned, which is very significant compared to 15,2% and 11,3% in Clusters 1 and 2, respectively. It is important to note that the variable Churn was excluded from the K-Means clustering to prevent the algorithm from forming clusters based on this information; therefore, the high churn density in Cluster 0 suggests an underlying explanatory power in the remaining features. See Table 5 in the Appendix for the coefficients of the principal components used in the analysis.

Cluster	Non-Churned Customers	Churned Customers (%)	Total
Cluster 0	1.297 (25,0%)	1.272 (68,1%)	2.569
Cluster 1	1.988 (38,4%)	357 (19,1%)	2.345
Cluster 2	1.889 (36,5%)	240 (12,8%)	2.129
Total	5.174 (100%)	1.869 (100%)	7.043

Table 2: Distribution of non-churned and churned customers across clusters

4.3 Class Imbalance

Particular attention was given to the class imbalance of the target variable Churn, as observed in Section 4.2, where non-churned customers compose 73.5% of the total sample. Most classification

algorithms perform optimally only when the number of samples in each class is roughly the same (Preuss 2025). This is especially important in this case, since the main objective is to correctly classify churned customers, and unbalanced models tend to be biased toward the majority class. Highly skewed datasets, where the minority class is heavily outnumbered by one or more classes, have proven to be a challenge while becoming increasingly common (Preuss 2025).

Several resampling strategies exist to address class imbalance, including random over-sampling, random under-sampling, and synthetic over-sampling methods. Chawla et al. 2002 introduced SMOTE (Synthetic Minority Over-sampling Technique) as an alternative that generates new synthetic samples of the minority class rather than replicating existing ones. The method works by selecting a minority class instance and one of its nearest neighbors, then creating new examples along the line segment joining the two in feature space. Given the limited sample amount of this dataset, it was preferable not losing potential useful information. SMOTE was chosen as an effective and computationally efficient method to balance the target variable Churn without reducing the number of majority class examples.

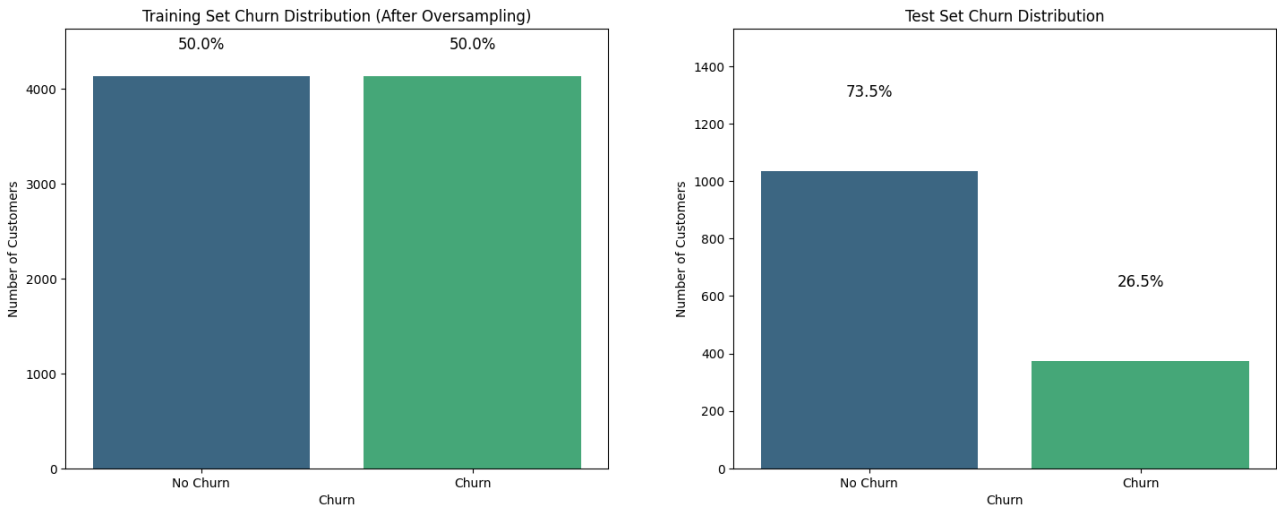


Figure 6: Churn distribution in the training and test set after applying SMOTE to the training set

Additionally, data leakage is an inherent risk when balancing imbalanced data. It occurs when information from the test set is inadvertently used during model training, leading to overly optimistic performance estimates. Sampling techniques are commonly used to address this imbalance; however, their implementation sometimes precedes the train–test split, potentially introducing data leakage (Kabane 2024). The test set was balanced after performing the 80–20 train–test split, and as shown in Figure 6, after oversampling the test set, it is now balanced while the training set maintains its original structure to perserve real-life distribution of the target variable.

4.4 Model Setup

Model selection focuses on presenting the classification algorithms used to model the data, where all models are trained on the SMOTE-adjusted training set and evaluated on the untouched test set. Logistic Regression serves as the baseline model due to its interpretability and computational efficiency. It is well suited for binary classification tasks (Géron 2017), estimating the log-odds of churn as a linear function of the input features while providing well-calibrated probability outputs. A vanilla configuration with `max_iter = 1000` is used to ensure convergence given the dimensionality of the dataset.

Ensemble methods are included as more flexible and non-linear models capable of capturing complex interactions in the data that linear models may miss. Random Forest is a bagging-based ensemble that constructs multiple decision trees on bootstrap samples and aggregates their predictions to improve generalisation and reduce variance (Breiman 2001). The classifier is configured with `n_estimators = 1000`, `max_depth = 10`, and `max_features = 'sqrt'`, using entropy as the splitting criterion and a mild class-weight adjustment to improve sensitivity to churned customers with depth constraints and minimum sample requirements help reduce overfitting.

In parallel, XGBoost is employed as a boosted-tree model that iteratively corrects the residual errors of previous trees, offering strong performance on structured data and robustness to class imbalance through built-in regularisation. The XGBoost configuration is selected via a small grid search, a hyperparameter optimisation technique that evaluates all possible combinations of selected parameters using cross-validation (Géron 2017). The parameters included in the grid were `n_estimators`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`, and `scale_pos_weight` again to prevent overfitting and improve sensitivity to churned customers.

4.5 Evaluation metrics

Aligned with the use case outlined in Section 3, the primary objective of this project is to accurately classify churned customers, minimizing cases incorrectly predicted as non-churned. Recall (particularly for the churned class) was selected as the main evaluation metric, as it reflects the model's ability to correctly identify churned customers and is directly affected by false negatives (see Formula 1). This metric was balanced with the F_1 -score, since precision and recall involve a trade-off: increasing recall typically leads to more false positives, thereby reducing precision, and vice versa. As observed in Formula 2, the F_1 -score is the harmonic mean of precision and recall, requiring both metrics to be high. It was therefore used to monitor this trade-off and mitigate excessive false positives resulting from recall maximization.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (1)$$

$$F_1 = \frac{2}{\text{Recall}^{-1} + \text{Precision}^{-1}} \quad (2)$$

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Predictions}} \quad (3)$$

Accuracy was also used to evaluate the models, ensuring that the overall correctness of the predictions was assessed in addition to the model’s ability to detect churned customers. Finally, ROC AUC was employed as a secondary metric to evaluate the model’s discriminative ability across all classification thresholds. As introduced by Hanley 1982 in his trial, “the area under the ROC curve corresponds to the probability of correctly identifying which of the two stimuli is ‘noise’ and which is ‘signal plus noise’”. It measures how well the model identifies positive and negative classes by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) (see Formula 4).

$$\text{ROC AUC} = \int_0^1 \text{TPR}(\text{FPR}), d\text{FPR} \quad (4)$$

5 Results

5.1 Model Performance and Selection

Table 3 reports the macro-averaged evaluation metrics for the three models, obtained by averaging performance across classes. The macro average is appropriate for imbalanced datasets because it weights each class equally, thereby providing a more informative assessment of model behaviour on the minority churn class. As observed in Table 3, the performance of the three models is similar.

Model	Recall	F_1 -score	Accuracy	Precision
Logistic Regression	0.82	0.80	0.84	0.80
Random Forest	0.84	0.80	0.83	0.79
XGBoost	0.83	0.81	0.84	0.80

Table 3: Overall evaluation metrics by model

The models discriminative ability is also similar, as indicated by the ROC area under the curve (AUC) scores presented in Figure 16 in the Appendix. These scores range from 0.919 to 0.923, indicating that all models achieve strong separation between churned and non-churned customers. Values

close to 1 reflect the model’s ability to maintain a high TPR while keeping the FPR low across all possible classification thresholds, thereby demonstrating robust threshold-independent discriminative performance of all three models.

Since the three models exhibit broadly similar macro-level performance and the primary objective of this study is to identify churned customers, the class-specific metrics for the churned class are decisive. Table 4 presents the class-specific metrics of the Random Forest, and the corresponding tables for the other models are shown in Tables 6 and 8 in the appendix. Among the three models, the Random Forest achieves the highest class-specific recall for churned customers, with a value of 0.89 compared to 0.76 for Logistic Regression and 0.80 for XGBoost. This means that the Random Forest model correctly identifies 89% of all churned customers in the test dataset, outperforming the other two models by a substantial margin.

Random Forest	Precision	Recall	F1-score	Support
Non-Churn (0)	0.95	0.80	0.87	1035
Churn (1)	0.62	0.89	0.73	374
Accuracy	-	-	0.83	1409
Macro avg	0.79	0.84	0.80	1409
Weighted avg	0.86	0.83	0.83	1409

Table 4: Classification report for Random Forest on the test dataset

The precision of the Random Forest is approximately 5–6 percentage points lower than that of the other models, and the confusion matrices in Figures 13, 14, and 15 in the appendix show that it produces more false positives. However, the model attains the same F_1 -score as XGBoost and a higher score than Logistic Regression. This indicates that the gain in recall (capturing a larger share of true churners) is balanced by only a moderate reduction in precision. Overall, the Random Forest improves the detection of churned customers without disproportionately increasing false positives, resulting in a well-calibrated trade-off between sensitivity and precision and making it the most suitable model given the objectives of this project.

5.2 Revenue at Risk (RaR) and Feature Importance

In this section, the objective is to estimate the probability that non-churned customers may churn in the future using the probabilities computed by the Random Forest model presented in Section 5. As outlined in Section 2, Revenue at Risk (RaR) translates these churn probabilities into a financial risk measure analogous to expected loss in credit risk modelling. Since only monthly charges are available, annual exposure is approximated as $12 \times \text{MonthlyCharges}_i$. The RaR measure is therefore defined as:

$$\text{RaR}_i = \hat{P}(\text{churn}_i) \times 12 \times \text{MonthlyCharges}_i \quad (5)$$

The churn probability $\hat{P}(\text{churn}_i)$ is obtained from the Random Forest model using the `predict_proba()` function. For each observation, every tree routes the customer through its learned splits to a terminal leaf node, where the class distribution of training samples provides the tree-level probability estimate. The final probability is the average of these per-tree estimates, yielding the ensemble-based prediction $\hat{P}(y = 1 \mid X_i)$. As shown in Figure 7, actual non-churned and churned customers exhibit distinctly different probability distributions, with mean values of 0.233 and 0.812, respectively. Notably, no churned customers appear below the non-churned probability mean (0.233), while many non-churned customers fall into this low-risk region, but the converse pattern is not observed.

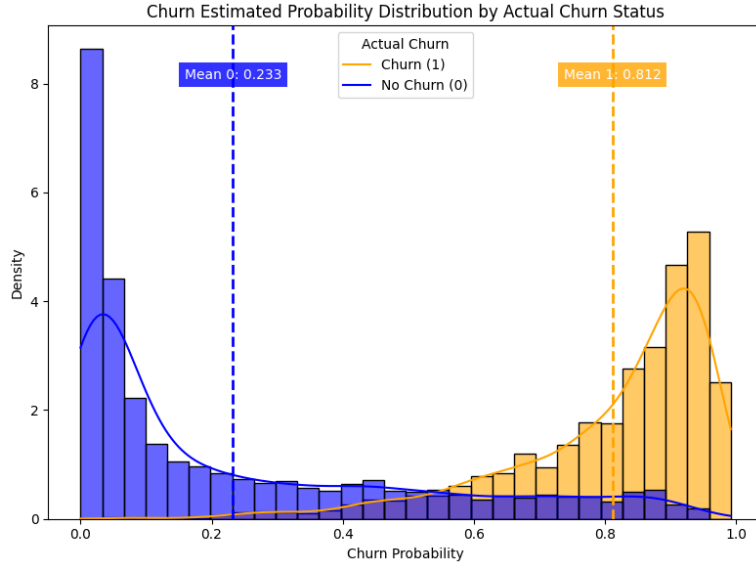


Figure 7: Estimated churn probabilities computed using the random forest classifier.

Using this definition, the RaR can be quantified by aggregating the annualised charges of non-churned customers whose predicted churn probability exceeds the 0.5 threshold. 969 customers fall are at high-risk risk and its total annual charges associated with these amount to \$743.722,20, out of a total portfolio value of \$3.803.829,00. This corresponds to 19,55% of the firm's expected annual revenue being at risk due to customers for whom churn is estimated to be more likely than not.

As explained in Section 2, interpretability is also a factor in ensuring transparency and trust in model outputs, and one common method to add an explainability layer is SHapley Additive exPlanations (SHAP), which assigns each feature an importance value for a particular prediction (Lundberg and Lee 2017). Using this approach, it becomes possible to identify which features contribute most to the model and whether they influence the outcome positively or negatively (in this case, the Churn

variable). The graphs in Figure 8 display the SHAP summary plot and mean absolute SHAP values, which together show both how individual feature values push churn predictions up or down (left) and the overall magnitude of each feature’s contribution to the model (right).

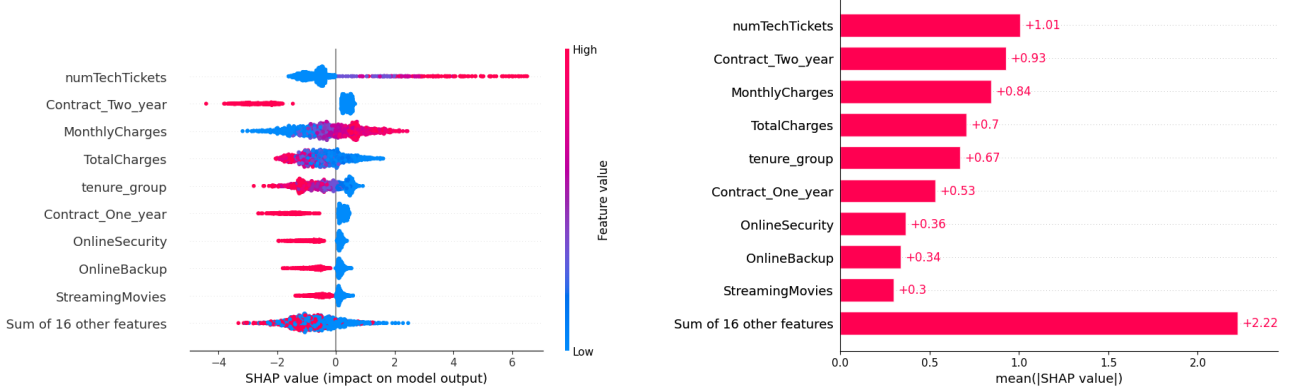


Figure 8: SHAP Feature importances and impact on Churn value

As observed in Figure 8, the number of technical support tickets opened (`numTechTickets`) is the feature with the strongest impact, consistently increasing the probability of churn. For the charge-related features (`MonthlyCharges` and `TotalCharges`), higher payment amounts are clearly associated with a higher probability of churn. Additionally, contract duration also plays an important role, as seen in `Contract_One_Year` and `Contract_Two_Year`: the longer the contract, the less likely a customer is to churn. Finally, `tenure_group` also shows notable importance and affects the model negatively, meaning that new customers are more likely to churn, whereas customers with higher tenure are less likely to do so.

6 Discussion

6.1 Suitability and Limitations

The selected models are well suited to the classification because they capture non-linear relationships between features and provide probabilistic outputs useful for the RaR computation, directly support managerial objective of identifying and quantifying the impact high-risk customers. However, the suitability of these models depends on several underlying assumptions, including the stationarity of customer behaviour over time, the presence of key determinants of churn in the dataset, and the validity of the synthetic minority samples introduced. If these assumptions do not hold, model predictions may fail to generalise, leading to biased results.

Several limitations arise from both data characteristics and modelling choices. Class imbalance remains a structural challenge, and although SMOTE balancing partially mitigates it, the approach

may oversimplify minority-class patterns or introduce artificial structure not present in real churn behaviour. Temporal drift constitutes another limitation, as shifts in pricing, competitive dynamics, or customer expectations can weaken model validity if not continually monitored. Moreover, metrics such as accuracy and ROC-AUC may hide poor model performance on the minority class, and an insufficient emphasis on recall may result in high-risk customers being systematically misclassified. Finally, the ensemble models applied in this study offer predictive strength at the cost of reduced interpretability, which may hinder managerial understanding of the drivers of churn.

6.2 Model Risk Management

The churn prediction model introduces several data-related risks that affect the reliability of its outputs and the resulting RaR estimates. A primary concern is data drift, which may arise when customer behaviour evolves due to changes in pricing structures, service offerings, or broader market conditions, causing the model to rely on patterns that no longer hold. Shifts in feature distributions as the customer base changes can also introduce feature bias, affecting the consistency of probability estimates. In addition, the use of synthetic samples during oversampling creates a dependency on interpolated minority-class patterns that may not fully capture the characteristics of churned customers. Mitigation requires routine drift detection, assessment of feature distribution stability, and continuous validation that the data environment remains aligned with the conditions under which the model was trained.

Model-intrinsic risks arise from the structure of the algorithms and from the interpretation of their outputs. Ensemble models may learn the training data too closely, increasing the risk of overfitting. This was mitigated through the use of regularisation techniques (see Section 4.4) and continuous monitoring of training and test performance to ensure that both remained aligned. Slightly better performance on the training data was expected and observed due to class balancing, but test results remained stable, indicating that the model did not overfit the synthetic minority samples. Balancing the dataset through SMOTE also partially addresses fairness concerns by improving minority-class representation, although the approach requires monitoring to ensure that synthetic samples do not introduce distortions. Evaluation choices can further contribute to model risk. Metrics such as accuracy or ROC AUC may obscure weak performance on churners, and insufficient emphasis on recall may limit the model's ability to support the managerial objective of identifying customers with elevated churn likelihood.

A structured model risk management and governance framework is therefore essential to ensure that predictions remain reliable after deployment. The modelling approach used in this study already addresses key AI governance pillars: robustness is supported through regularisation and cross-validated performance checks, fairness is enhanced by balancing the training data, and interpretability is strengthened through SHAP values and clustering-based segmentation. Transitioning such a model into production requires clearly defined accountability for model use, comprehensive documentation

of modelling choices and assumptions, and traceability mechanisms that enable retrospective examination of predictions. Ongoing governance involves periodic backtesting on out-of-time samples, monitoring for feature and concept drift, recalibrating probabilities where necessary, and maintaining transparent documentation. Overall, the framework used contributes significantly to reduce organisational risk by improving the identification and quantification of at-risk customers while managing the risks implied by the use of machine learning.

7 Conclusion

This study aimed to model customer churn as a risk event and to quantify its financial impact through the RaR framework. Fine-tuned ensemble models (Random Forest and XGBoost) were trained alongside logistic regression as a baseline, with the Random Forest model producing the strongest results, achieving a churned class-specific recall of 0.89 with accuracy comparable to the other models. Applying its predicted probabilities indicated that 19.55% of the customer portfolio is at elevated churn risk, corresponding to an annualised exposure of \$743,722.20. In parallel, K-Means clustering identified a customer segment containing 68.1% percent of all churned customers, and key churn drivers were found to relate to customer charges, technical support interactions, contract length, and customer seniority.

Overall, the modelling framework strengthens the organisation's ability to identify at-risk customers and quantify potential revenue loss, thereby contributing directly to churn-related risk reduction. The analysis also identified and mitigated key data, model and governance risks, including exposure to data drift, overfitting, and constraints on interpretability, all of which require continued monitoring and structured oversight. When accompanied by these safeguards, the risk reduction achieved through improved identification of churn-prone customers offsets the AI-related risks implied by the modelling approach.

Appendix

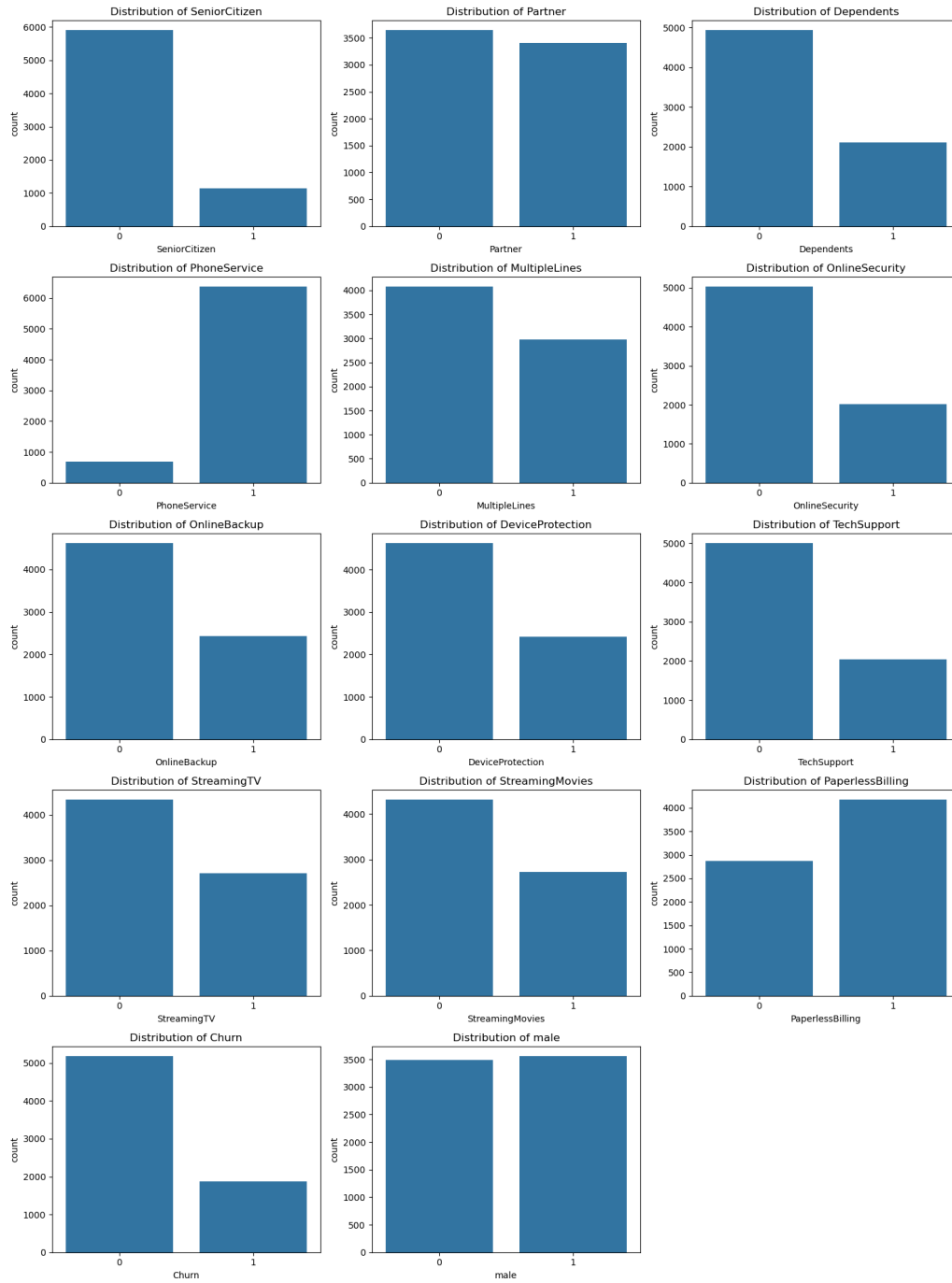


Figure 9: Histogram grid for all binary features of the data after preparation.

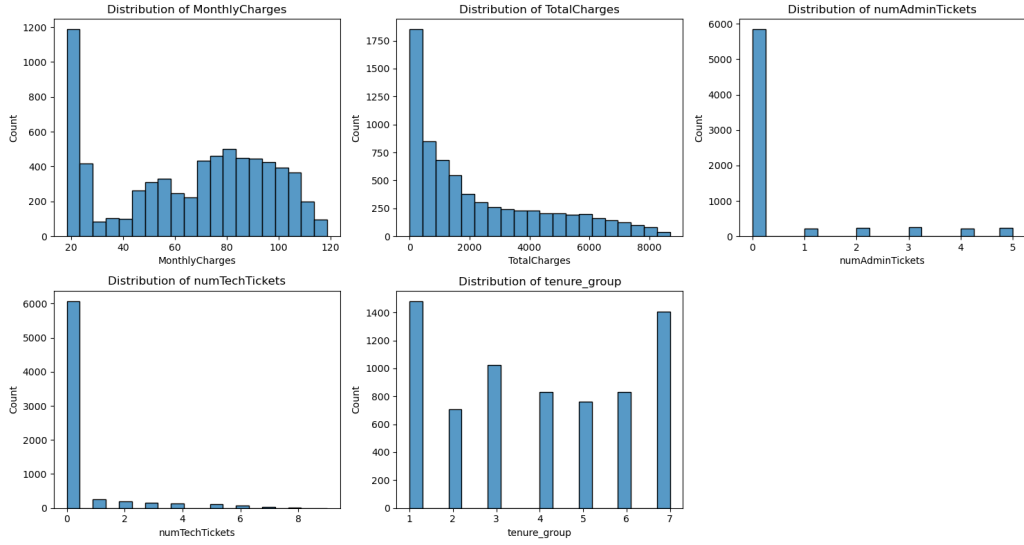


Figure 10: Histogram grid for all numerical and categorical features of the data after preparation.

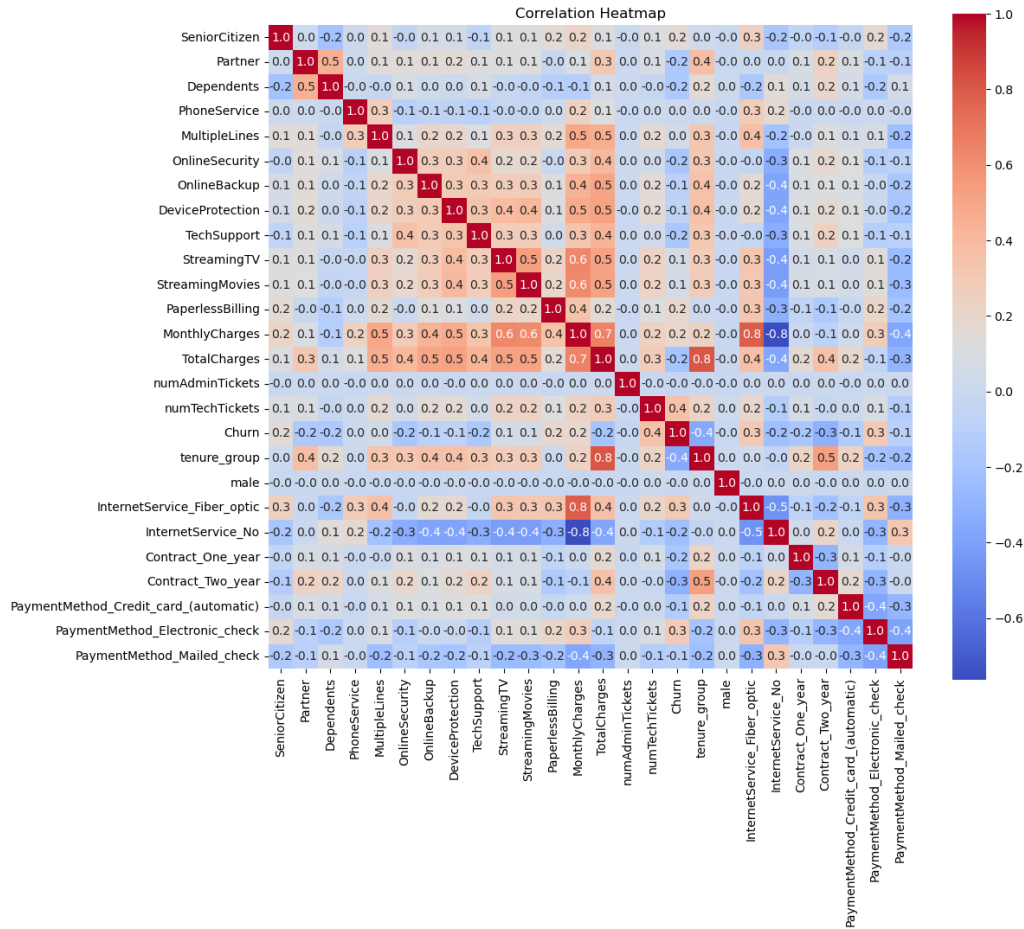


Figure 11: Correlation heatmap for all features of the data after preparation.

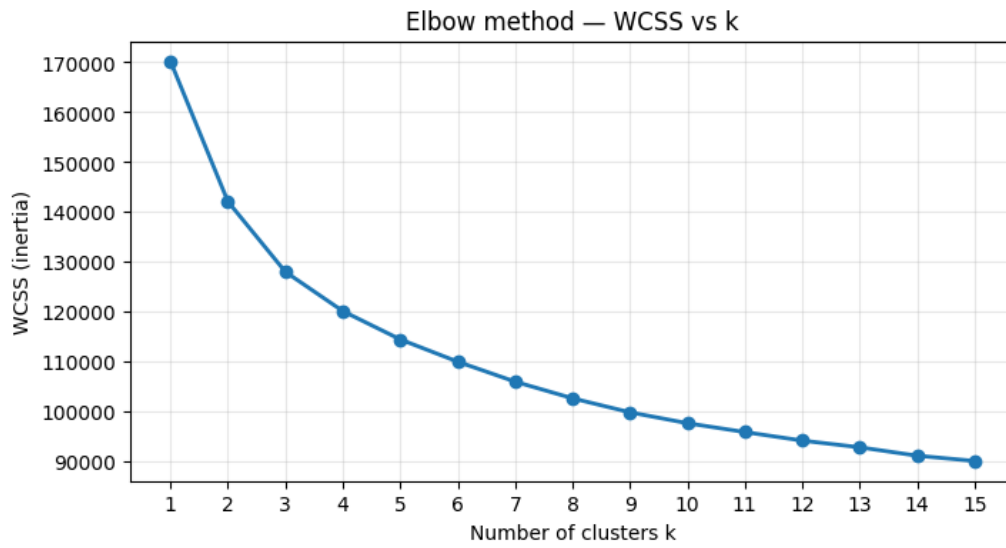


Figure 12: K-Means clustering performance across different values of k.

Variable	PC1	PC2
SeniorCitizen	0.091542	-0.191250
Partner	0.112471	0.249256
Dependents	-0.007899	0.272658
PhoneService	0.031805	-0.068244
MultipleLines	0.222392	-0.028533
OnlineSecurity	0.177063	0.193255
OnlineBackup	0.238402	0.083996
DeviceProtection	0.259370	0.093850
TechSupport	0.198226	0.190620
StreamingTV	0.287326	-0.041033
StreamingMovies	0.288401	-0.039018
PaperlessBilling	0.144483	-0.217015
MonthlyCharges	0.380131	-0.187433
TotalCharges	0.363520	0.187433
numAdminTickets	0.002829	0.002316
numTechTickets	0.144169	-0.026579
tenure_group	0.240936	0.347796
male	-0.005814	0.005033
InternetService_Fiber_optic	0.237685	-0.317307
InternetService_No	-0.228961	0.270266
Contract_One_year	0.041611	0.096834
Contract_Two_year	0.066646	0.382132
PaymentMethod_Credit_card_(automatic)	0.066800	0.204921
PaymentMethod_Electronic_check	0.071594	-0.370278
PaymentMethod_Mailed_check	-0.198073	0.107373

Table 5: Principal component coefficients for PC1 and PC2

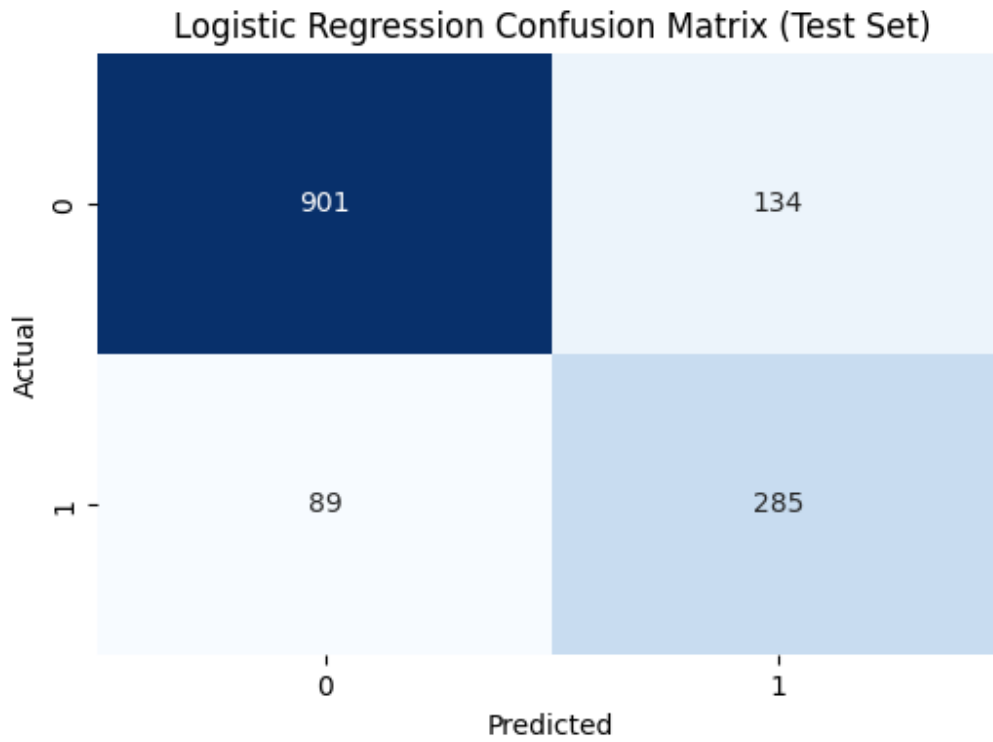


Figure 13: Confusion matrix for Logistic Regression on the test dataset

Logistic Regression	Precision	Recall	F1-score	Support
Non-Churn (0)	0.91	0.87	0.89	1035
Churn (1)	0.68	0.76	0.72	374
Accuracy	-	-	0.84	1409
Macro avg	0.80	0.82	0.80	1409
Weighted avg	0.85	0.84	0.84	1409

Table 6: Classification report for Logistic Regression on the test dataset

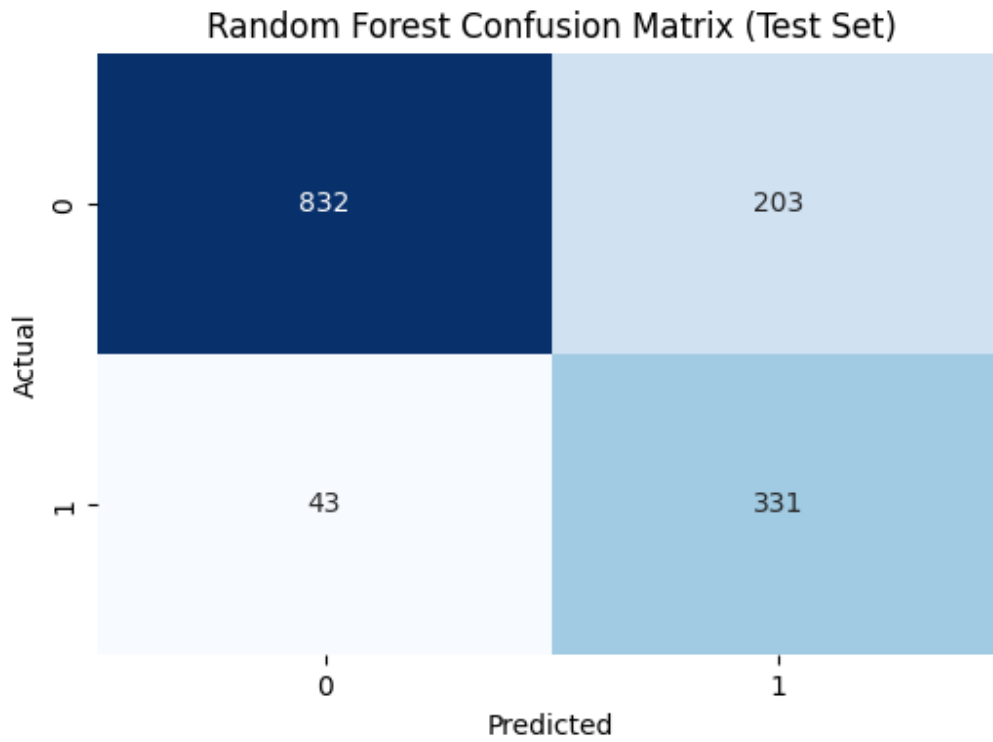


Figure 14: Confusion matrix for Random Forest on the test dataset

Random Forest	Precision	Recall	F1-score	Support
Non-Churn (0)	0.95	0.80	0.87	1035
Churn (1)	0.62	0.89	0.73	374
Accuracy	-	-	0.83	1409
Macro avg	0.79	0.84	0.80	1409
Weighted avg	0.86	0.83	0.83	1409

Table 7: Classification report for Random Forest on the test dataset

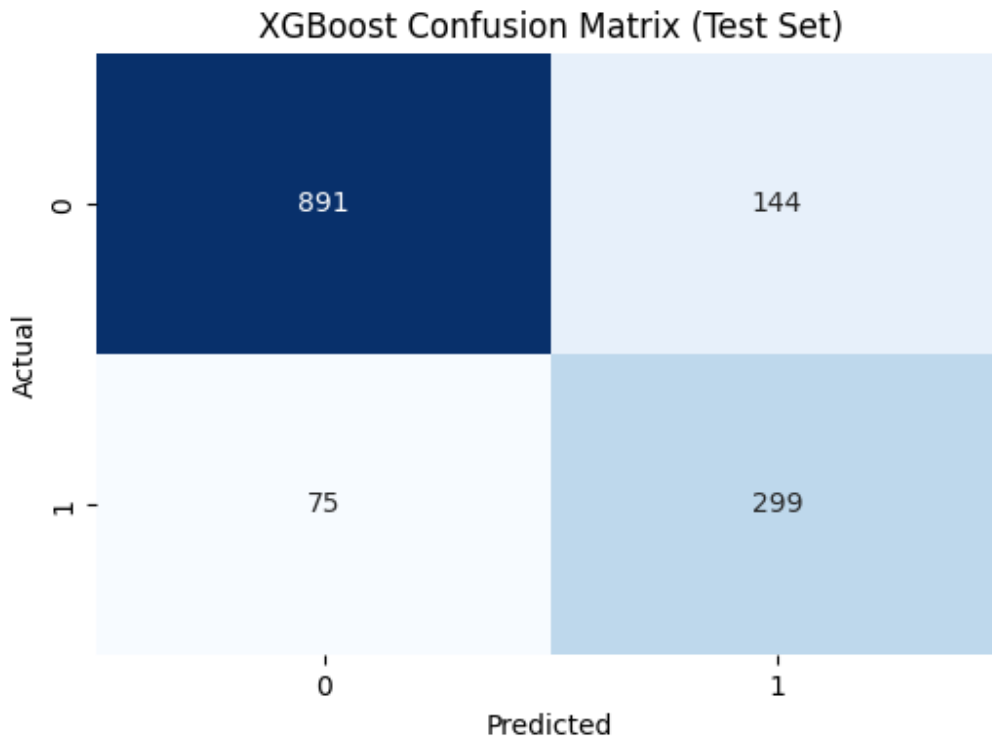


Figure 15: Confusion matrix for XGBoost on the test dataset

XGBoost	Precision	Recall	F1-score	Support
Non-Churn (0)	0.92	0.86	0.89	1035
Churn (1)	0.67	0.80	0.73	374
Accuracy	-	-	0.84	1409
Macro avg	0.80	0.83	0.81	1409
Weighted avg	0.86	0.84	0.85	1409

Table 8: Classification report for XGBoost on the test dataset

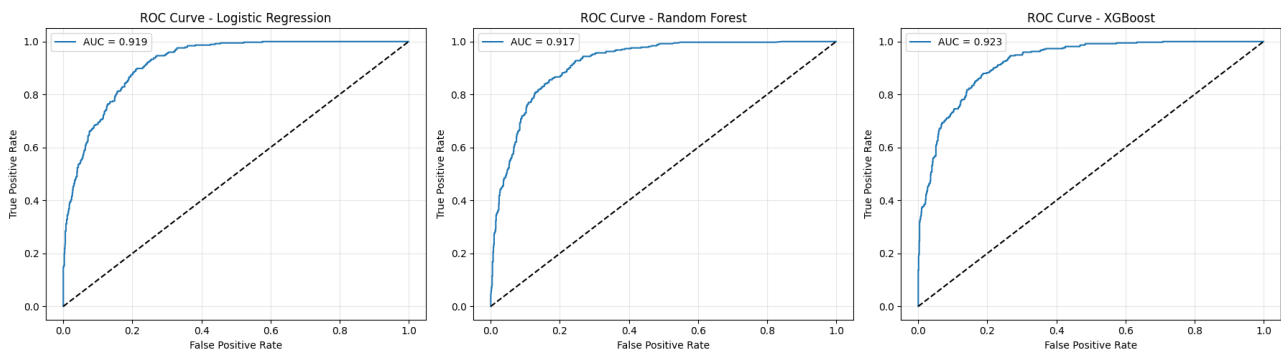


Figure 16: ROC curve and the Area Under the Curve (AUC) for each model.

References

- Basel Committee on Banking Supervision (2005). *International Convergence of Capital Measurement and Capital Standards: A Revised Framework*. Tech. rep. Basel: Bank for International Settlements.
- Breiman, Leo (2001). “Random Forests”. In: *Machine Learning* 45.1. DOI: 10 . 1023 / a : 1010933404324. URL: <https://doi.org/10.1023/a:1010933404324>.
- Chawla, Nitesh V. et al. (2002). “SMOTE: Synthetic Minority Over-sampling Technique”. In: *Journal of Artificial Intelligence Research*. DOI: 10 . 1613/jair . 953. URL: <https://doi.org/10.1613/jair.953>.
- Esty, Benjamin and Daniel Fisher (Apr. 2019). “MoviePass: The ”Get Big Fast” Strategy”. In: *Harvard Business School Articles*.
- Géron, A. (2017). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*. 2nd ed. O’Reilly Media.
- Hanley, J.A. (May 1982). “The Meaning and Use of the Area Under a Receiver Operating Characteristic (ROC) Curve”. In: *Radiology* 143. DOI: 10.1148/radiology.143.1.7063747.
- Hansson, Sven Ove (Jan. 2004). “Philosophical Perspectives on Risk”. In: *Techné: Research in Philosophy and Technology* 8. DOI: 10 . 5840/techne2004818.
- Kabane, Siyaxolisa (2024). *Impact of Sampling Techniques and Data Leakage on XGBoost Performance in Credit Card Fraud Detection*. arXiv: 2412 . 07437 [cs.LG]. URL: <https://arxiv.org/abs/2412.07437>.
- Lundberg, Scott M. and Su-In Lee (2017). “A Unified Approach to Interpreting Model Predictions”. In: URL: <https://arxiv.org/abs/1705.07874>.
- Neslin, Scott et al. (Apr. 2006). “Defection Detection: Measuring and Understanding the Predictive Accuracy of Customer Churn Models”. In: *Journal of Marketing Research American Marketing Association ISSN* 43. DOI: 10 . 1509/jmkr . 43 . 2 . 204.
- Peng, Ke, Yan Peng, and Wenguang Li (Dec. 2023). “Research on customer churn prediction and model interpretability analysis”. In: *PLOS ONE* 18. DOI: 10 . 1371/ journal . pone . 0289724. URL: <https://doi.org/10.1371/journal.pone.0289724>.
- Preuss, Bjørn (2025). *Lecture 6: Unbalanced Datasets*. Lecture slides, Copenhagen Business School. Quantitative Risk Management – A Machine Learning Approach.